



Operations Research

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Exercise Sheet 6

Demo Exercises

Exercise D6.1. *Branch-and-Bound*

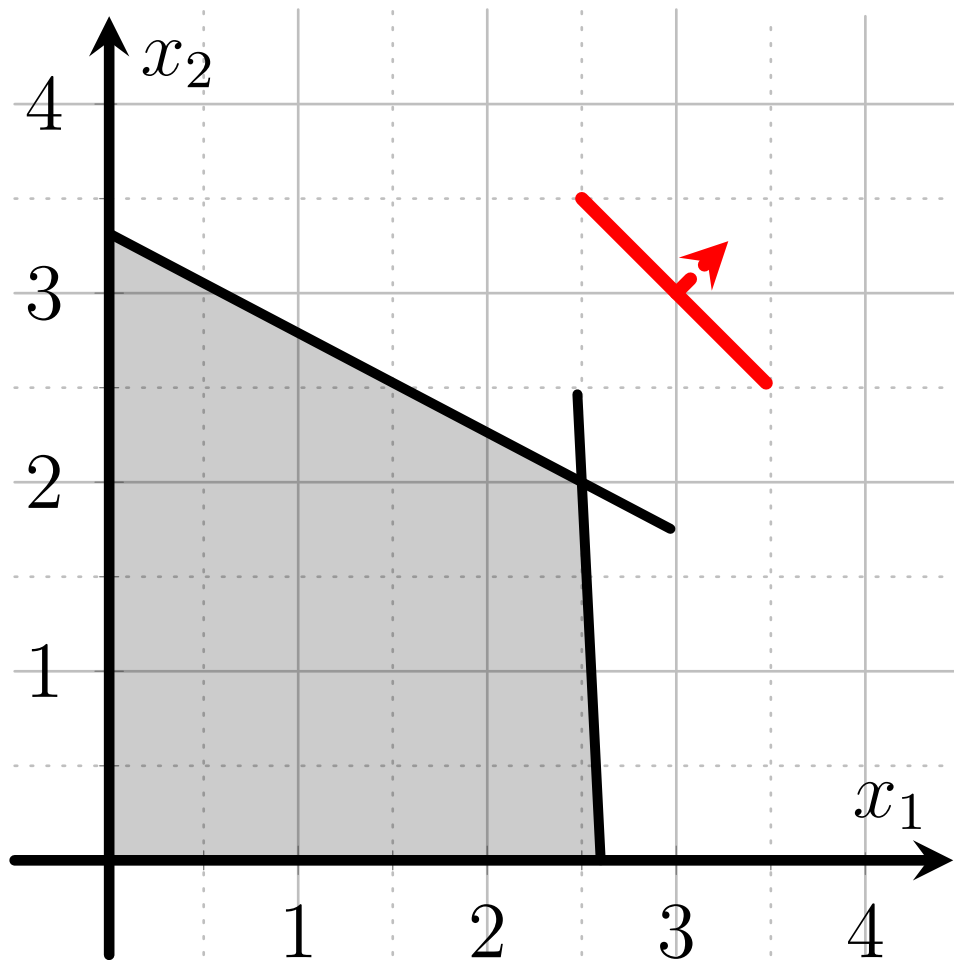
Consider the following *maximization problem*, given as an integer program (P)

$$\begin{aligned} (P) \quad & \underset{\text{s.t.}}{\text{Max}} && x_1 + x_2 \\ & 10x_1 + 19x_2 &\leq & 63 \\ & 10x_1 + 0.5x_2 &\leq & 26 \\ & x_1, x_2 &\in & \mathbb{N}_0 \end{aligned}$$

Solve the integer linear program (P) using the branch-and-bound method. Specify the optimal values for x_1 and x_2 , as well as the optimal value of the objective function.

At each node, label the "greater-or-equal" branch with a lower subscript and the "less-than-or-equal" branch with a higher subscript. Process subproblems in first-in-first-out *FIFO* order.

Note: In the following coordinate system, the feasible set of LP relaxation of (P) and the objective function are shown. You can use it to solve the subproblems graphically.



Exercise D6.2. *Cutting planes*

Given the following IP:

$$\begin{aligned}
 (P) \quad & \text{Max}_{\text{s.t.}} \quad 4x_1 + 2x_2 \\
 & 2x_1 + 3x_2 \leq 3 \\
 & -2x_1 - 2x_2 \geq -5 \\
 & x_1, x_2 \in \mathbb{N}_0
 \end{aligned}$$

- Give the LP relaxation of (P) in canonical form.
- Solve the LP relaxation using the simplex method and give the optimal solution and the optimal objective function value.
- Give the inequality of a possible Gomory cut with respect to the determined optimal tableau.

Exercise D6.3. *Cut the Planes*

The aircraft manufacturer *Airtruck* has developed two new models, the *Airtruck A381* and the *Airtruck A382*. Since production relies on limited resources, the company must decide how many units of each model to build in order to maximise profit.

This decision is modelled with the following integer programme:

$$\max z = x_1 + x_2 \quad (1)$$

$$\text{s.t. } x_1 + 2x_2 + s_1 = 5 \quad (2)$$

$$3x_1 + x_2 + s_2 = 6 \quad (3)$$

$$x_1 + s_3 = 2.4 \quad (4)$$

$$x_2 + s_4 = 3.8 \quad (5)$$

$$s_1, s_2, s_3, s_4 \geq 0 \quad (6)$$

$$x_1, x_2 \in \mathbb{N}_0 \quad (7)$$

After solving the LP relaxation with the simplex method, the optimal tableau is:

Base	Coefficients						Res.
	x_1	x_2	s_1	s_2	s_3	s_4	
z	0	0	$2/5$	$1/5$	0	0	$16/5$
x_1	1	0	$-1/5$	$2/5$	0	0	$7/5$
x_2	0	1	$3/5$	$-1/5$	0	0	$9/5$
s_3	0	0	$1/5$	$-2/5$	1	0	1
s_4	0	0	$-3/5$	$1/5$	0	1	2

- Identify the basic variables that can be used to generate Gomory fractional cuts, and explain why they are eligible.
- The company tests both Gomory cuts. Derive **all** valid representations of **each** Gomory cut from the tableau above.
- Simplify the Gomory cut obtained from the x_1 -row so that it can be directly incorporated into the integer programme (use the form without slack variables).
- Formulate the updated integer programme including the first Gomory cut. You may reference earlier equations by their number (e.g., (1)) instead of rewriting them.